

CS3001 – Computer Networks

Quiz – 4B

Name	
Roll. No.	
Section	

CLO1: Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.

Part 1 [4 Marks]: A TCP sender behaves as follows: **On timeout:** next timeout is doubled (exponential backoff). **On ACK or app data arrival:** recompute TimeoutInterval from EstimatedRTT and DevRTT. Ignore RTT samples for any retransmitted transmission.

Initial state at $t = 0$:

- EstimatedRTT = 150 ms, DevRTT = 30 ms $\rightarrow$ TimeoutInterval = 270 ms.
- Segment S0 sent at $t = 0$.
- Use $\alpha = 0.20$ and $\beta = 0.30$.

Events:

- $t = 270$ ms: timeout; S0 is retransmitted; next timeout set to 540 ms (no recompute).
- $t = 420$ ms: ACK for original S0 arrives (use only this ACK for SampleRTT; ignore retransmission path).

Compute EstimatedRTT, DevRTT, TimeoutInterval immediately after the ACK at 530 ms.

Part 1 [6 Marks]: Assume MSS = 1 segment. Use:

- On timeout: ssthresh = floor(cwnd/2) (using cwnd just before loss), then cwnd = 1.
- On triple-duplicate ACK: ssthresh = floor(cwnd/2), then cwnd = ssthresh.

Events occur at RTT boundaries (loss at end of that RTT) Start: cwnd = 1, ssthresh = 12.

- RTT1: no loss.
- RTT2: no loss.
- RTT3: no loss.
- RTT4: loss detected by triple-duplicate ACK.
- RTT5: no loss.
- RTT6: loss detected by timeout.
- RTT7: no loss.

Fill in cwnd and ssthresh at the start of each RTT in the following tabular manner:

Event	cwnd value	ssthresh value
<Event>	<Integer value>	<Integer value>

Part 1:

1. $\text{SampleRTT} = 420 \text{ ms} - 0 \text{ ms} = 420 \text{ ms}$
2. $\text{EstimatedRTT1} = (1 - \alpha) \times \text{EstimatedRTT0} + \alpha \times \text{SampleRTT}$
 $= 0.8 \times 150 + 0.2 \times 420$
 $= 120 + 84$
 $= 204 \text{ ms}$
3. $\text{DevRTT1} = (1 - \beta) \times \text{DevRTT0} + \beta \times |\text{SampleRTT} - \text{EstimatedRTT1}|$
 $= 0.7 \times 30 + 0.3 \times |420 - 204|$
 $= 21 + 0.3 \times 216$
 $= 21 + 64.8$
 $= 85.8 \text{ ms}$
4. $\text{Timeout1} = \text{EstimatedRTT1} + 4 \times \text{DevRTT1}$
 $= 204 + 4 \times 85.8$
 $= 204 + 343.2$
 $= 547.2 \text{ ms} \rightarrow \text{rounded to } \approx 547 \text{ ms}$

Checking at 530ms does not change anything. The only ACK that matters occurred at 420ms (from the question: *use only this ACK for SampleRTT*).

Part 2:

Step-by-step:

- Start RTT1: $\text{cwnd}=1, \text{sssthresh}=12$
- Start RTT2: slow start $\rightarrow \text{cwnd}=2$
- Start RTT3: slow start $\rightarrow \text{cwnd}=4$
- Start RTT4: slow start $\rightarrow \text{cwnd}=8$
- End of RTT4: triple-dup $\rightarrow \text{sssthresh}=\text{floor}(8/2)=4, \text{cwnd}=4$
- Start RTT5: $\text{cwnd}=4, \text{sssthresh}=4$ (now in congestion avoidance)
- Start RTT6: linear increase $\rightarrow \text{cwnd}=5, \text{sssthresh}=4$
- End of RTT6: timeout $\rightarrow \text{sssthresh}=\text{floor}(5/2)=2, \text{cwnd}=1$
- Start RTT7: $\text{cwnd}=1, \text{sssthresh}=2$ (slow start resumes)